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JSS Mahavidyapeetha
JSS Science And Technology University
(Established Under JSS Science and Technology University Act No. 43 of 2013)
(Formerly Known as SJCE)

JSS SCIENCE AND TECHNOLOGY UNIVERSITY MYSURU

III Semester BE CIE-2/Event-2

Branch: CS&E **Section: B**

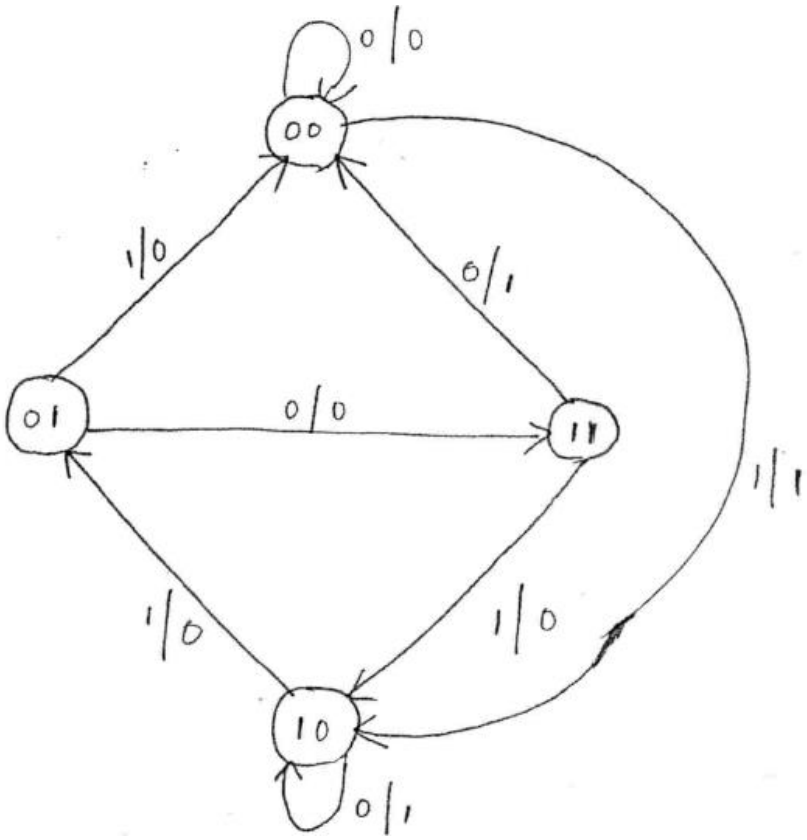
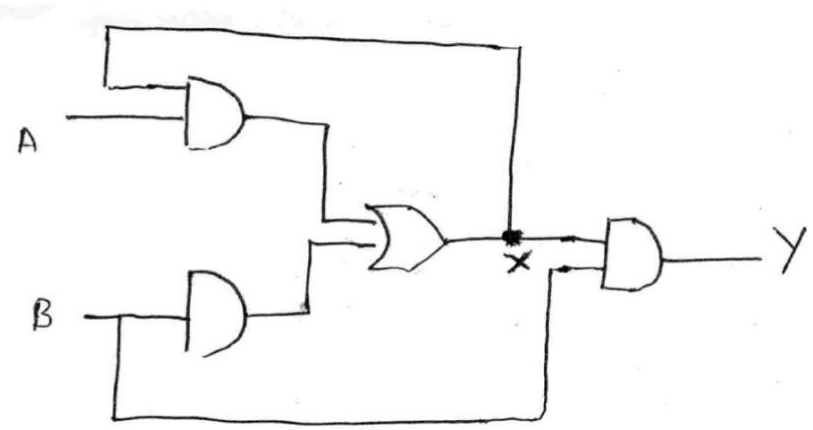
Name of the Paper Setter: Prof. KCS **CODE NO: 20CS330**

Digital System Design

Submission Date: 25-02-2022 **Max. Marks: 20**

Course Outcome:	
CO–1	Understand the fundamental concepts of Boolean algebra, Boolean theorems, K-MAP, Quine-McCluskey algorithm and combinational and sequential circuits.
CO–2	Apply the knowledge of theoretical foundations to simplify and solve Boolean functions
CO–3	Using the knowledge of Boolean algebra and behaviour, timing aspects, operation of various flip-flops analyze the combinational and sequential circuits.
CO–4	Design and implement combinational circuits, synchronous/asynchronous counters and register circuits.
CO–5	Evaluate the performance of sequential circuits using various models

Q. No	CO	Cognitive Domain	Question	Marks
1	CO-5	Application	Design synchronous counter using clocked D flip flops for the given counting sequence: 0->20>3->6->5->1->0.	5m
2	CO-5	Knowledge	Difference between Moore and Mealy machine along with block diagram.	5m
3	CO-5	Application	A sequential circuit has one input and one output. The state diagram is shown below figure. Implement the sequential circuit using T flipflop.	5m

				
4	CO-5	Analysis/ Application	Derive the transition table and state diagram for the asynchronous sequential circuit shown in below figure: 	5m

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